Question 1: Risk Assessment Scoring

An auditor is evaluating a data center's external risk factors. The auditor assigns scores to various risk components as follows:

- **Proximity to flood zones:** 8/10
- **Crime rate in the area:** 7/10
- **Historical power outage frequency:** 6/10
- **Proximity to industrial facilities:** 5/10

If each factor is equally weighted, calculate the overall external risk score for the data center.

Solution:

To calculate the overall external risk score, sum the individual scores and divide by the number of factors:

Overall Risk Score=(8+7+6+5) / 4= 26 / 4 = 6.5

Therefore, the overall external risk score is **6.5 out of 10**.

Question 2: Annual Loss Expectancy (ALE) Calculation

A data center is located in an area prone to earthquakes. The following data is available:

- **Single Loss Expectancy (SLE):** $500,000
- **Annual Rate of Occurrence (ARO):** 0.02

Calculate the Annual Loss Expectancy (ALE) for earthquakes affecting the data center.

Solution:

The ALE is calculated using the formula:

ALE=SLE×ARO

Plugging in the values:

ALE=$500,000×0.02=$10,000

Therefore, the Annual Loss Expectancy due to earthquakes is **$10,000**.

Question 3: Cost-Benefit Analysis of Security Measures

To mitigate the risk of power outages, a data center considers installing an uninterruptible power supply (UPS) system costing $150,000. The system is expected to reduce the impact of outages,

lowering the Single Loss Expectancy (SLE) from $200,000 to $50,000. The Annual Rate of Occurrence (ARO) for power outages is 0.05. Calculate the Return on Investment (ROI) for the UPS system over one year.

Solution:

First, calculate the Annual Loss Expectancy (ALE) before and after implementing the UPS system:

- **Before UPS:**

  $ALE_{before} = \$200,000 \times 0.05 = \$10,000$

- **After UPS:**

  $ALE_{after} = \$50,000 \times 0.05 = \$2,500$

Next, determine the annual cost savings:

$$\text{Annual Savings} = ALE_{before} - ALE_{after} = \$10,000 - \$2,500 = \$7,500$$

Finally, calculate the ROI:

$$\begin{aligned} \text{ROI} &= (\text{Annual Savings} - \text{Cost of UPS}) / \text{Cost of UPS} \times 100 \\ &= (\$7,500 - \$150,000) / \$150,000 \times 100 = -95\% \end{aligned}$$

Therefore, the ROI for the UPS system over one year is **-95%**, indicating that the investment would not be recouped within a year.

Question 4: Calculating Downtime Impact

A data center experiences an average of 3 power outages per year, each lasting approximately 2 hours. The estimated cost of downtime is $10,000 per hour. Calculate the total annual cost of downtime due to power outages.

Solution:

First, calculate the total downtime hours per year:

$$\text{Total Downtime} = \text{Number of Outages} \times \text{Duration per Outage} = 3 \times 2 = 6 \text{ hours}$$

Next, calculate the total annual cost:

$$\text{Total Annual Cost} = \text{Total Downtime} \times \text{Cost per Hour} = 6 \times \$10,000 = \$60,000$$

Therefore, the total annual cost of downtime due to power outages is **$60,000**.

Question 5: Evaluating Redundancy Investment

A data center is considering investing in a redundant internet connection to mitigate the risk of ISP failures. The current Annual Loss Expectancy (ALE) due to internet outages is $25,000. The redundant connection costs $15,000 annually and is expected to reduce the ALE to $5,000. Determine whether the investment is financially justified over one year.

Solution:

Calculate the annual cost savings from the redundant connection:

$\text{Annual Savings} = \text{ALE}_{\text{current}} - \text{ALE}_{\text{with redundancy}} = \$25{,}000 - \$5{,}000 = \$20{,}000$

Compare the annual savings to the cost of the redundant connection